In this assessment, students will learn about using Docker Swarm. The goal is to understand general concepts and behaviors related to containers orchestration and rollout.

1. Starting Docker Swarm on local VM

By default Swarm is disabled. You can check that behavior using the command *docker info:*

```
guest@guest-VirtualBox:~/ex_7$ docker info
Client:
 Context:    default
 Debug Mode: false
 Plugins:
  app: Docker App (Docker Inc., v0.9.1-beta3)
  buildx: Build with BuildKit (Docker Inc., v0.5.1-docker)

Server:
 Containers: 29
  Running: 2
  Paused: 0
  Stopped: 27
 Images: 110
 Server Version: 20.10.6
 Storage Driver: overlay2
  Backing Filesystem: extfs
  Supports d_type: true
  Native Overlay Diff: true
  userxattr: false
 Logging Driver: json-file
 Cgroup Driver: cgroupfs
 Cgroup Version: 1
 Plugins:
  Volume: local
  Network: bridge host ipvlan macvlan null overlay
  Log: awslogs fluentd gcplogs gelf journald json-file local logentries splunk syslog
 Swarm: inactive
 Runtimes: io.containerd.runc.v2 io.containerd.runtime.v1.linux runc
```

To start swarm, you can execute the following command:

```
guest@guest-VirtualBox:~/ex_7$ docker swarm init
Swarm initialized: current node (v21u3xehnkdmodsniycbr7xn9) is now a manager.

To add a worker to this swarm, run the following command:

    docker swarm join --token SWMTKN-1-0ragih0gcssum8w0zmfu8kuzumt8usfca2fvn2js2xwquf9bxk-5nr30ww5eoejf3e208c6bh7d4
10.0.2.15:2377

To add a manager to this swarm, run 'docker swarm join-token manager' and follow the instructions.
```

As can be seen in the above image, a new swarm cluster was initialized with 1 node. The token can be used to add other machines to that swarm cluster.
To check the health of the nodes you can execute the following command:

```
guest@guest-VirtualBox:~/ex_7$ docker node ls
ID                            HOSTNAME            STATUS    AVAILABILITY   MANAGER STATUS   ENGINE VERSION
v21u3xehnkdmodsniycbr7xn9 *   guest-VirtualBox    Ready     Active         Leader           20.10.6
```

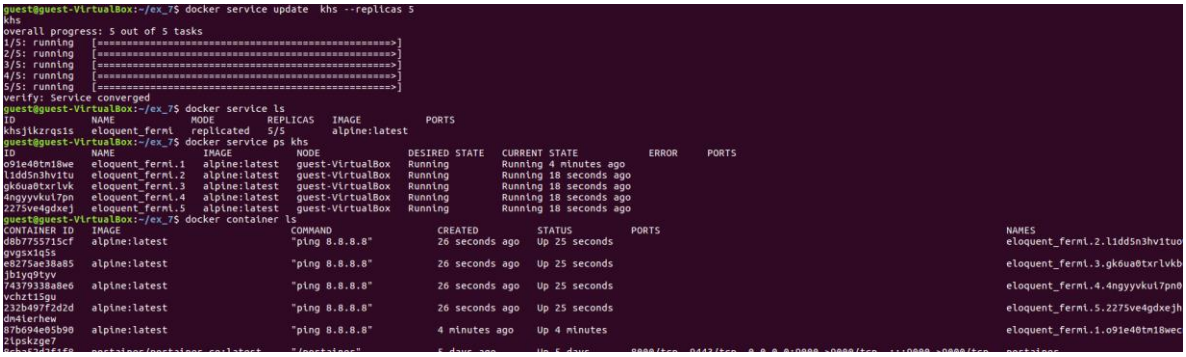
2. Starting a basic service on the cluster

We will now run a simple service on the cluster. For this we will create a new service that ping the Google DNS using alpine image.
E.g.:

```
guest@guest-VirtualBox:~/ex_7$ docker service create alpine ping 8.8.8.8
khsjikzrqs1s6urkenevspvhv
overall progress: 1 out of 1 tasks
1/1: running   [==================================================>]
verify: Service converged
guest@guest-VirtualBox:~/ex_7$ docker service ls
ID             NAME             MODE         REPLICAS   IMAGE          PORTS
khsjikzrqs1s   eloquent_fermi   replicated   1/1        alpine:latest
guest@guest-VirtualBox:~/ex_7$ docker service ps khs
ID             NAME               IMAGE          NODE              DESIRED STATE   CURRENT STATE            ERROR     PORTS
o91e40tm18we   eloquent_fermi.1   alpine:latest   guest-VirtualBox   Running         Running 48 seconds ago
```

2.1 – Scale up the service to 5 replicas.

Hint: The following image describes how you can scale up the service through the command line:



2.2- What are the typical scenarios where you want to scale up the service?

2.3- Try to remove a container from that service using a command similar to:

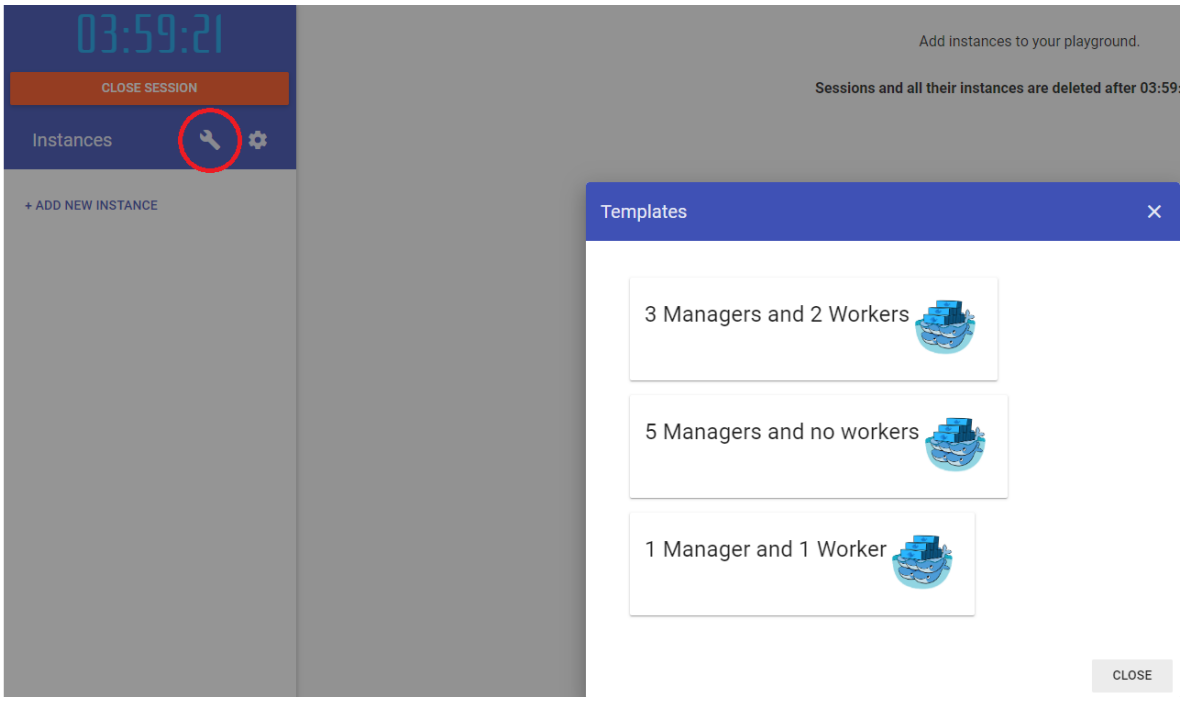
$ docker container rm –f <container_id>

What will happen to the cluster? It will have only 4 replicas?

2.4 – Remove the containers from the cluster, but now use the command “docker service rm <service name>

3. Using Docker Swarm on 5 machines

Docker Swarm makes more sense when we have a distributed cluster among several machines. To test this scenario, we will use an online tool that give us several machines – during 4 hours.
The student can access Docker Playground (https://labs.play-with-docker.com/). Afterwards, you can add 5 instances (3 managers and 2 workers) to the playground using the provided templates:

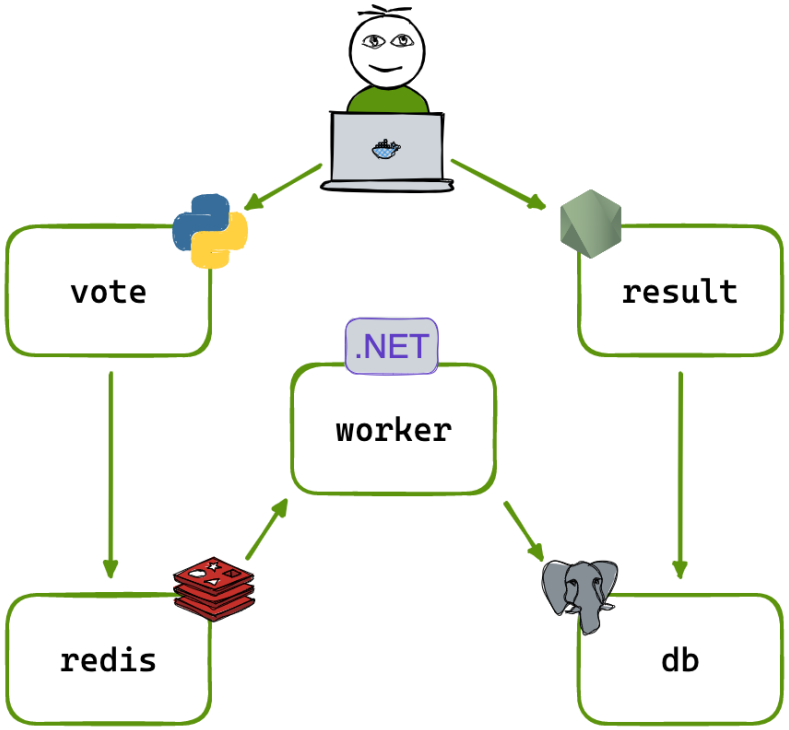


3.1 – Can you execute the command “docker node ls” on a machine that is only a worker?

3.2 – Try to create the same mentioned service on the playground (“docker service create –replicas 3 alpine ping 8.8.8.8”). How are the replicas distributed among the cluster?

4. Docker Stack

We are going to deploy a more complex application using docker stack. The application has the following flow according to here:



Try to start the application on the docker playground using the swarm command mentioned on github – e.g. *docker stack deploy --compose-file docker-stack.yml vote*

Check that the application is working correctly (e.g. vote app on port 8081, results on 8080) and what kind of distribution was made among the nodes.